Teaching survival analysis to clinical collaborators

Emily C. Zabor
Department of Quantitative Health Sciences

Joint Statistical Meetings
August 3, 2026

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Why is education on survival analysis needed?

Survival analysis is common, especially in cancer research

Overall survival, progression-free survival, and time to relapse are examples of the time-to-event endpoints that form the foundation of oncology trials and observational studies.

Both the need for and interpretation of survival analyses are misunderstood

Collaborators, reviewers, and readers all routinely misjudge both why a survival analysis method is needed and what the results mean.

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Overall survival is not the only endpoint

- Time to death
- Time to progression
- Time to hospital readmission
- Time to heart attack
- Time to local recurrence
- Time to transplant
- Time to metastasis

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Censoring drives the need for survival analysis

500 patients

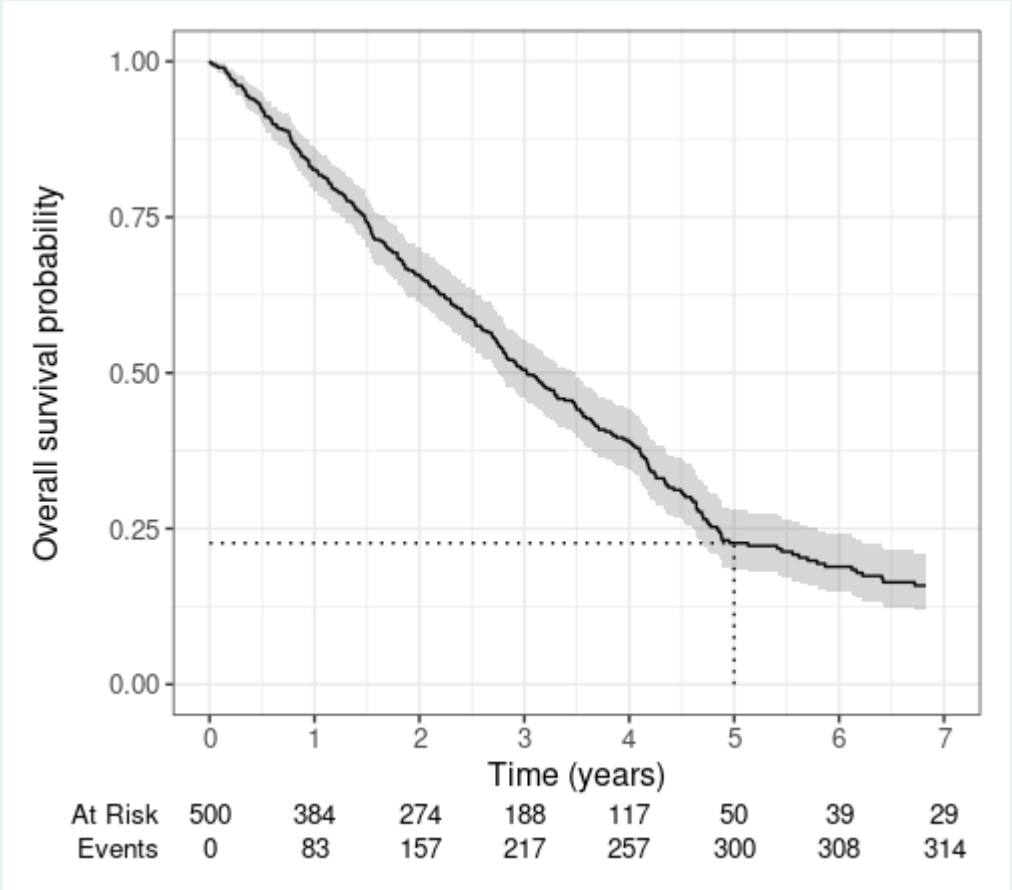
300 die by 5 years

Crude 5-year probability of survival:

$$\left(1 - \frac{300}{500}\right) \times 100 = 40\%$$

Ignores the fact that 150 patients were censored prior to 5 years, so their status as alive or dead at 5 years is not known.

Kaplan-Meier 5-year probability of survival = 23%



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Incorrect interpretations are common

INCORRECT

“The hazard ratio was 0.3 for treated versus untreated, which means there was a 70% reduced risk of death on treatment.”

CORRECT

“The hazard ratio was 0.3 for treated versus untreated, which means there was a 70% reduced hazard of death on treatment at any moment during follow-up.”

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Modes of education

Formal

Structured, classroom-based

Graduate students

(heard about this in James's talk)

Medical researchers

(up next in Nathan's talk)

Informal

Ad hoc, on demand

Non-classroom based

Journal clubs, study reports, and everyday conversations

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Opportunities for informal education

1

Journal clubs

Use published articles to walk through methods and interpretation together as a group

2

Study reports with detailed interpretations

Write results sections that explain what the numbers mean in detail, with words

3

Verbally when in-person

Ask questions early, and take opportunities to explain the need for survival analysis in real time

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Choosing what to teach in journal clubs

- Select articles on a clinical topic that matches your audience's interest
- Build a series across sessions to cover different methods in order of complexity

Always include an example of an error in application or interpretation

Real mistakes make a bigger impact than abstract rules

A sample series

1. **Kaplan–Meier method**
2. **Cox regression**
3. **Time-dependent covariate / landmark analysis**

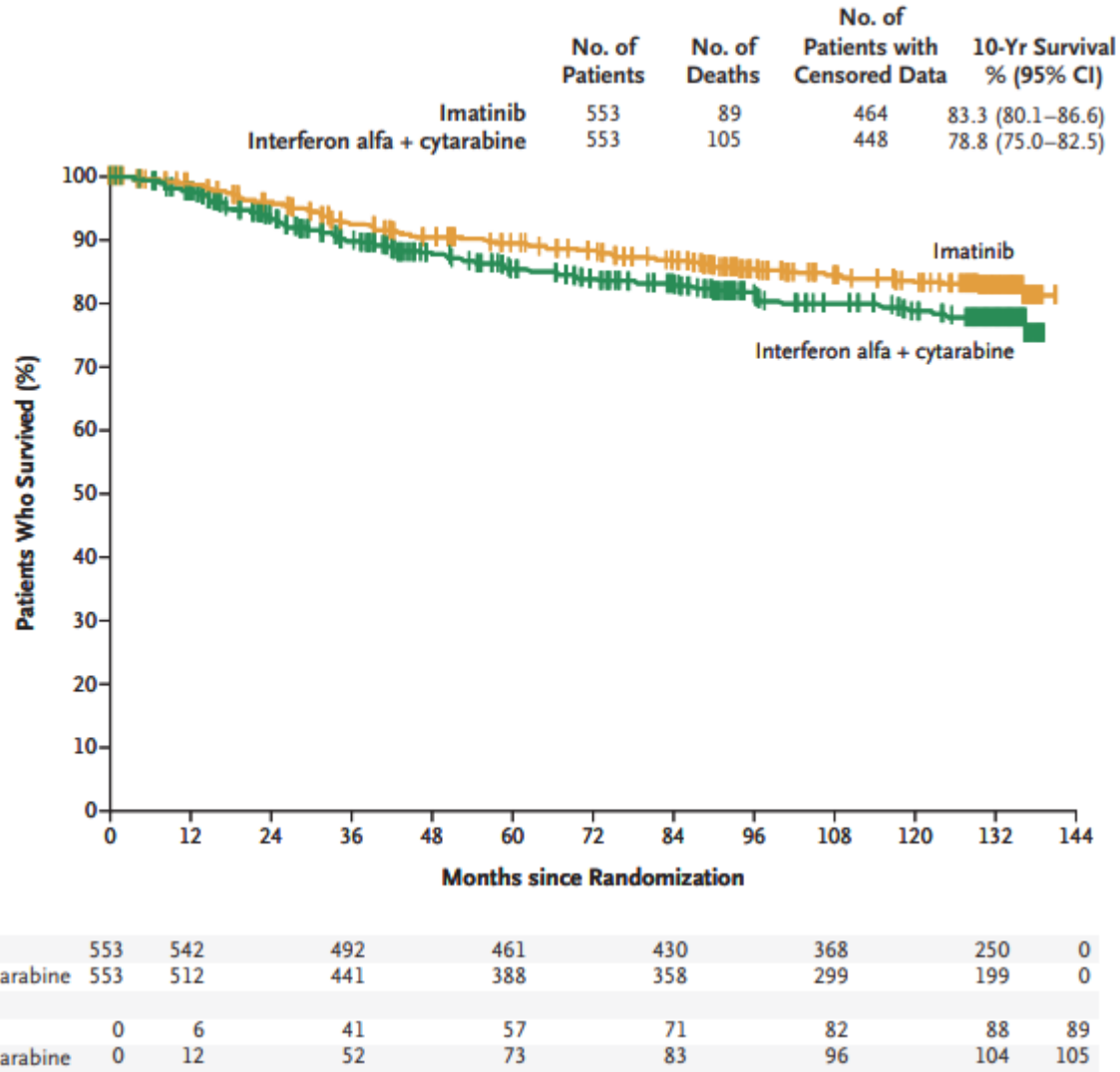
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The NEW ENGLAND JOURNAL of MEDICINE

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- Good use of ***risk table + censoring ticks***
- Shows number of patients and number of deaths – ***not percentage of deaths***
- Demonstrates ***administrative censoring*** at the end of follow-up

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Important not to interpret this as a mortality rate

20.1% of patients were censored and follow-up time varied, as noted in text

Table 2. Deaths during the Trial among Patients Randomly Assigned to Imatinib.

Variable	Patients (N = 553)
	no. (%)
Total deaths	89 (16.1)
Cause of death	
Chronic myeloid leukemia	
Without hematopoietic stem-cell transplantation	37 (6.7)
After hematopoietic stem-cell transplantation	13 (2.4)
Secondary malignant condition*	11 (2.0)
Cardiac disorder or cardiovascular disease	7 (1.3)
Infectious disease	5 (0.9)
Other	16 (2.9)

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"Patients who died or who had data censored before each landmark analysis were excluded from that landmark analysis."

Shows the number included and number of events in each landmark analysis subset alongside results

Table 4. Landmark Analysis of Outcomes at 10 Years According to Molecular Response Levels at 12 Months and 18 Months in Patients Treated with First-Line Imatinib Therapy Who Could Be Evaluated.*

Variable	Major Molecular Response or Better	Lack of Major Molecular Response	P Value
At 12 mo			
No. of patients who could be evaluated	153	151	
Death — no. (%)	15 (9.8)	22 (14.6)	
Not related to CML	11 (7.2)	7 (4.6)	
Related to CML	4 (2.6)	15 (9.9)	
Estimated 10-yr overall survival — % (95% CI)	91.1 (86.5–95.7)	85.3 (79.5–91.1)	0.15
Estimated 10-yr freedom from CML-related death — % (95% CI)	97.8 (95.4–100)	89.4 (84.3–94.5)	0.007
At 18 mo			
No. of patients who could be evaluated	164	89	
Death — no. (%)	12 (7.3)	13 (14.6)	
Not related to CML	12 (7.3)	4 (4.5)	
Related to CML	0	9 (10.1)	
Estimated 10-yr overall survival — % (95% CI)	93.0 (89.0–97.0)	85.6 (77.9–93.2)	0.04
Estimated 10-yr freedom from CML-related death — % (95% CI)	100 (100–100)	90.5 (84.1–96.8)	<0.001

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Include interpretations alongside results

Don’t

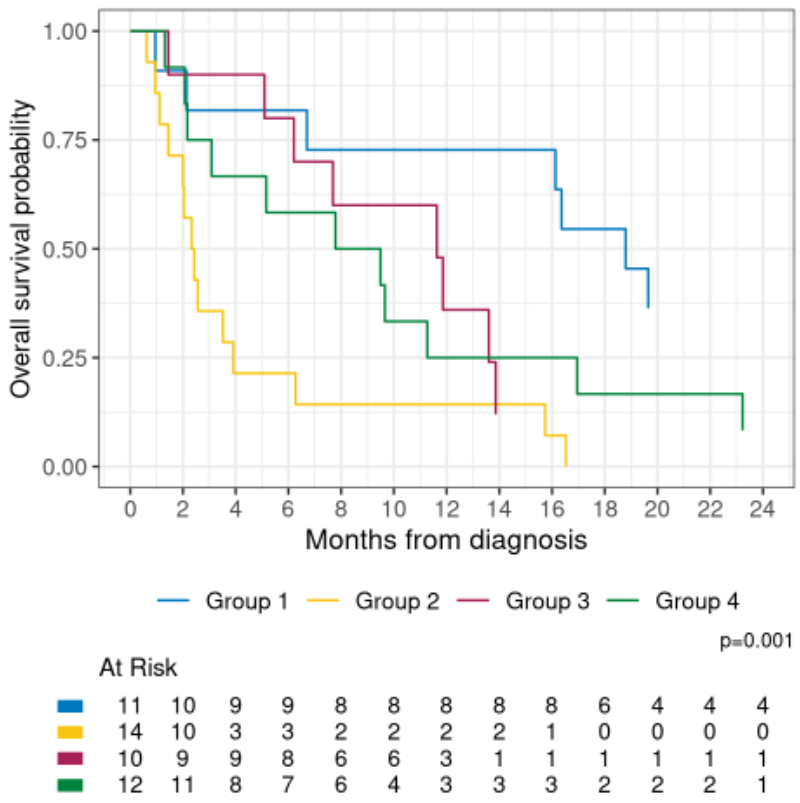
Present a KM curve or hazard ratio with no explanation of what it is or what it means

Do

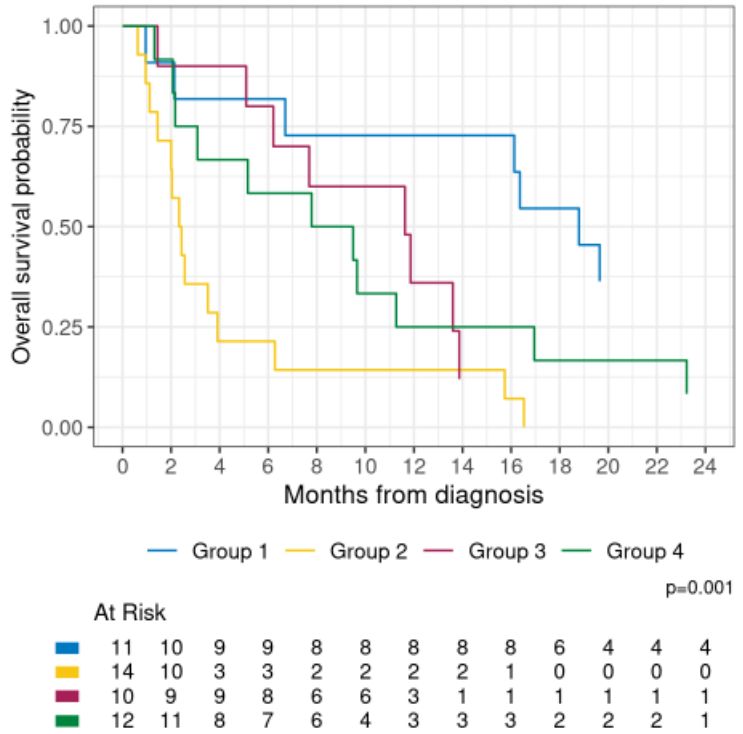
Alongside every result, say what the method is and what the result indicates, in language that you would find a peer-reviewed clinical journal

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My project report without details



My project report with details



Overall survival probability was estimated using the Kaplan-Meier method. There is a significant difference in overall survival probability by group according to the log-rank test (p=0.001). Group 2 has the lowest overall survival probability. Note the small sample size and few at risk as time from diagnosis increases, leading to greater uncertainty at the later timepoints.

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Ask questions early and often

1 Ask about time components

Is there a natural time-to-event structure, even if no one has framed it that way?

2 Ask what data are available

Diagnosis dates, death dates, scan dates, last contact dates – what is captured, and in what format?

3 Explain why it matters

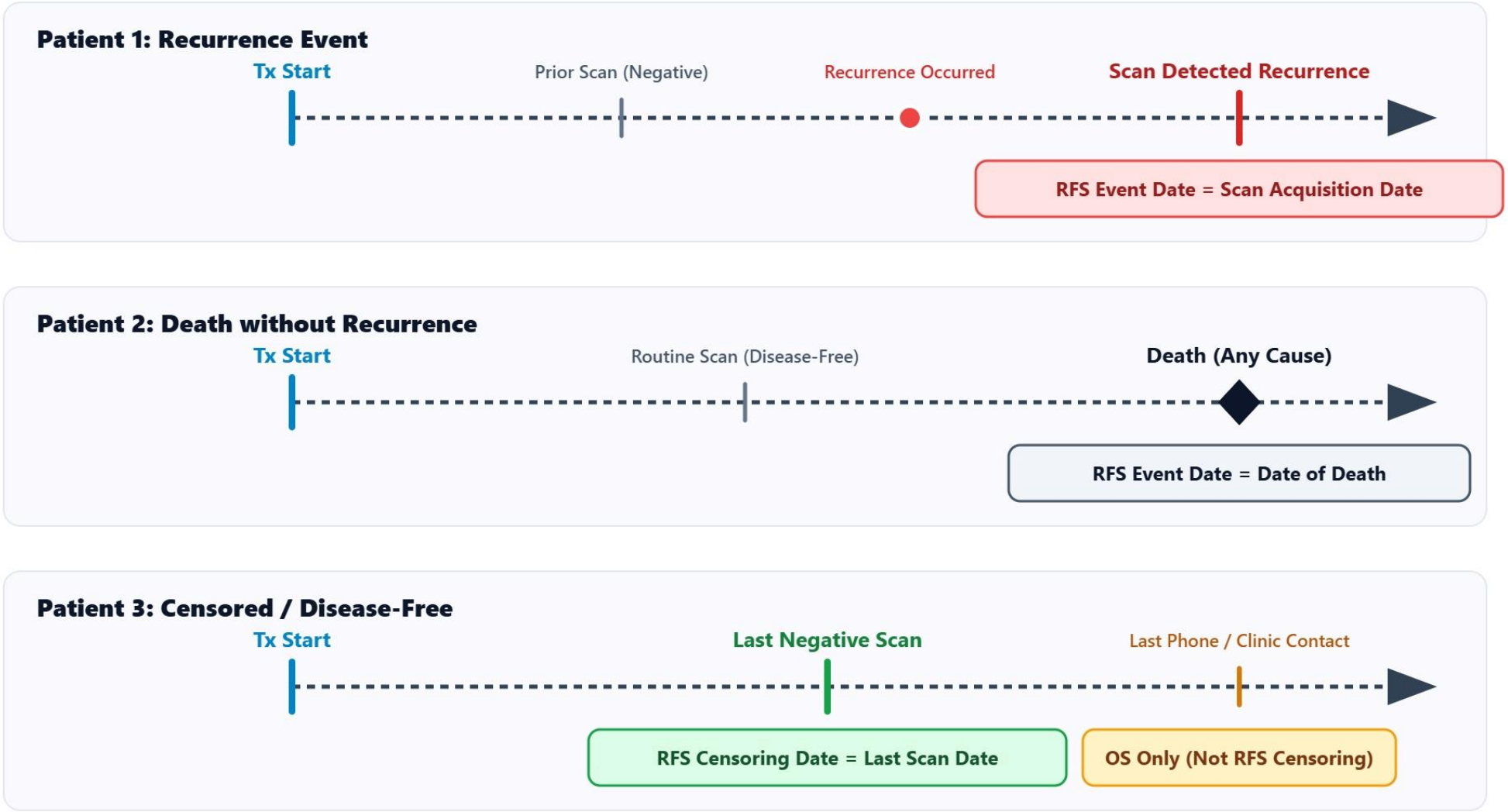
Ignoring time components or handling time incorrectly can produce biased results.

4 Give guidance on what to collect

Needed dates are not always obvious, discuss endpoints and list dates needed to calculate times for each.

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Pictures or diagrams can help convey ideas



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Thank you! Questions?

zabore2@ccf.org

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